SEA4001W: Exercise 4 - Calculate the Taylor series expansion

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Exercise 4

general formula (3rd order)

1) $f(x) \approx f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)(x-x_0)^2}{2!} + \frac{f^{(3)}(x_0)(x-x_0)^3}{3!} + O\left((x-x_0)^4\right)$

slide 51

$f(x) = \frac{x}{x+1}$ * quotient rule $\frac{d}{dx}\left[\frac{u}{v}\right] = \frac{(v.u' - u.v')}{v^2}$, $\underbrace{u = x}$, $v = x+1$, $v^2 = (x+1)^2$

$u' \,\&\, v' = 1$ $\therefore = (x+1) - x = 1$

$\therefore f'(x) = \frac{(x+1)(1) - x(1)}{(x+1)^2} = \frac{1}{(x+1)^2}$

$\therefore f''(x) = \frac{-2}{(x+1)^3}$ → differentiate $f'(x)$ using the chain rule* $= -2(x+1)^{-3}.1 = \frac{-2}{(x+1)^3}$

$x = x_0$

$\therefore f(x_0) = \frac{x_0}{x_0+1}$ & $f'(x_0) = \frac{1}{(x_0+1)^2}$ & $f''(x_0) = \frac{-2}{(x_0+1)^3}$

$= f(x) = \frac{x_0}{x_0+1} + \frac{1}{(x_0+1)^2}(x-x_0) - \frac{1}{(x_0+1)^3}(x-x_0)^2 + O\left((x-x_0)^3\right)$

$\frac{f''(x_0)}{2!} = \frac{\frac{-2}{(x_0+1)^3}}{2} = \frac{-1}{(x_0+1)^3}$

halves the f'' term

2) $f(x) = \frac{2x^{\cancel{8}}}{x^{\cancel{8}2}} = \frac{2}{x^2} = 2x^{-2}$ → so that we don't have to use the quotient rule, can now use power rule

$f'(x) = 2.(-2).x^{-3}$ → $\frac{d}{dx}[ax^n] = \underbrace{nax^{n-1}}$, $n = -2$, $a = 2$ $= -2.2x^{-2-1} = \underline{-4x^{-3}}$

$= -4x^{-3}$

$= \frac{-4}{x^3}$

$f''(x) = -4.(-3).x^{-4}$ → $\frac{d}{dx}[ax^n] = \underbrace{nax^{n-1}}$, $n = -3$, $a = -4$ $= -3.-4x^{-3-1} = 12x^{-4}$

$= 12x^{-4} = \frac{12}{x^4}$

2 cont.) . $x = x_0$

$\therefore f(x_0) = \frac{2}{x_0}$ & $f'(x_0) = \frac{-4}{x_0^3}$ & $f''(x_0) = \frac{12}{x_0^4}$

$f(x) = \frac{-2}{x_0^2} - \frac{4}{x_0^3}(x - x_0) + \frac{6}{x_0^4}(x - x_0)^2 + O((x - x_0)^3)$

$$\frac{f''(x_0)}{2!} = \frac{\frac{12}{x_0^4}}{2} = \frac{6}{x_0^4}$$

3) $f(x) = \ln x^2$ $\quad \ln(x^n) = n\ln(x)$ (log power rule)

$= 2\ln(x)$

$f'(x) = 2 \cdot \frac{1}{x} = \frac{2}{x} = 2x^{-1}$ $\quad \frac{d}{dx}[\ln x] = \frac{1}{x}$

$f''(x) = \frac{d}{dx}[2x^{-1}] = -2x^{-2} = \frac{-2}{x^2}$

power rule $(-1) \cdot 2 \cdot x^{-2}$

$= -2x^{-2}$

$x = x_0$

$f(x_0) = 2\ln(x_0)$ & $f'(x_0) = \frac{2}{x_0}$ & $f''(x_0) = \frac{-2}{x_0^2}$

$\therefore f(x) = 2\ln(x_0) + \frac{2}{x_0}(x - x_0) - \frac{1}{x_0^2}(x - x_0)^2 + O((x - x_0)^3)$

$$\frac{f''(x_0)}{2!} = \frac{\frac{-2}{x_0^2}}{2} = \frac{-1}{x_0^2}$$